



## AIRCRAFT MAINTENANCE MANUAL

### MAIN HYDRAULIC SYSTEM – ADJUSTMENT/TEST

\*\*ON A/C ALL

#### **TASK 29-10-00-710-801**

#### **Operational Test of the No. 1 and No. 2 Hydraulic System Differential Pressure Indicator (DPI) and Case Drain Manifold Bypass Valve**

##### 1. General

- A. The maintenance procedure that follows is for the operational test of the No. 1 and No. 2 Hydraulic System Case Drain Filter Manifold Differential Pressure Indicator (DPI) and Bypass Valve (bypass valve).
- B. The test will make sure the case drain DPI and bypass valve operate correctly. It will make sure the DPI sets at the specified pressure and the bypass valve opens and closes at the specified pressures. If the bypass valve does not operate correctly, then it can be replaced.
- C. The maintenance procedure that follows is only applicable as follows:
  - For the No. 1 hydraulic system with filter manifolds that have part numbers 945169-3, 945169-5, or 945169-7.
  - For the No. 2 hydraulic system with filter manifolds that have part numbers 945169-3, 945169-5, and without Modsum 4-113545 and Pre-SB 84-29-21.

##### 2. Job Set-Up Information

###### Subtask 29-10-00-943-049

###### A. Tools and Equipment

- |     |                        |                           |
|-----|------------------------|---------------------------|
| (1) | Commercially available | Hydraulic fluid container |
|-----|------------------------|---------------------------|

###### Subtask 29-10-00-944-016

###### B. Consumable

- |     |                        |                        |
|-----|------------------------|------------------------|
| (1) | Commercially available | Wiper cloth, lint free |
|-----|------------------------|------------------------|

###### Subtask 29-10-00-946-030

###### C. Reference Information Table

| REFERENCE                             | DESIGNATION                                                             |
|---------------------------------------|-------------------------------------------------------------------------|
| <a href="#">TASK 12-10-29-210-801</a> | Fluid Quantity Check of the No. 1 and No. 2 Hydraulic System Reservoirs |
| <a href="#">TASK 12-10-29-611-801</a> | Servicing of the No. 1 and No. 2 Hydraulic System Reservoirs            |

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-10-00



**AIRCRAFT MAINTENANCE MANUAL**

| REFERENCE                             | DESIGNATION                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------|
| <a href="#">TASK 20-20-11-400-801</a> | Installation of Hydraulic Lines and Fittings                                       |
| <a href="#">TASK 29-00-00-863-801</a> | Pressurization of the No. 1 and No. 2 Hydraulic Systems                            |
| <a href="#">TASK 29-00-00-863-802</a> | Removal of the Hydraulic Pressure from the No. 1 and No. 2 Hydraulic Systems       |
| <a href="#">TASK 29-00-00-910-801</a> | Hydraulic Safety Precautions                                                       |
| <a href="#">TASK 29-10-00-870-801</a> | Bleeding of the No. 1 Hydraulic System                                             |
| <a href="#">TASK 29-10-00-870-802</a> | Bleeding of the No. 2 Hydraulic System                                             |
| <a href="#">TASK 29-11-31-000-801</a> | Removal of the Hydraulic Filter Manifold                                           |
| <a href="#">TASK 29-11-31-400-801</a> | Installation of the Hydraulic Filter Manifold                                      |
| <a href="#">TASK 29-11-34-000-801</a> | Removal of the No. 1 and No. 2 Case-Drain Filter Elements                          |
| <a href="#">TASK 29-11-34-400-801</a> | Installation of the No. 1 and No. 2 Case-Drain Filter Elements                     |
| <a href="#">TASK 29-11-37-000-801</a> | Removal of the No. 1 Hydraulic System Filter Manifold Case Drain Bypass Valve      |
| <a href="#">TASK 29-11-37-400-801</a> | Installation of the No. 1 Hydraulic System Filter Manifold Case Drain Bypass Valve |
| <a href="#">TASK 32-00-00-840-801</a> | Open and Lock the Landing Gear Doors                                               |
| <a href="#">TASK 32-00-00-840-802</a> | Unlock and Close the Landing Gear Doors                                            |

3. Job Set-Up

Subtask 29-10-00-910-026

- A. Obey the hydraulic safety precautions (Refer to [TASK 29-00-00-910-801](#)).

Subtask 29-10-00-010-047

**WARNING:** YOU MUST INSTALL THE LOCKPINS IN THE DOOR MECHANISMS OF THE MLG AND NLG. THE DOOR MECHANISMS CAN ACCIDENTALLY CLOSE THE LANDING GEAR DOORS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

- B. Open and lock the landing gear doors (Refer to [TASK 32-00-00-840-801](#)).



## AIRCRAFT MAINTENANCE MANUAL

Subtask 29-10-00-865-077

C. Open and tag the circuit breakers that follow:

| CB PANEL AND CB NO. | NAME           |
|---------------------|----------------|
| RIGHT DC, Q4        | STBY HYD PMP 1 |
| RIGHT DC, L8        | HYD PWR XFER   |

### 4. Procedure

**WARNING:** OBEY ALL THE HYDRAULIC SAFETY PRECAUTIONS WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM OR A HYDRAULIC SYSTEM COMPONENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

**WARNING:** DO NOT TOUCH THE DPI OR THE AREA AROUND IT WITH YOUR BARE FINGER. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURY TO YOURSELF. MAKE SURE YOU USE HEAT PROTECTIVE GLOVES.

Subtask 29-10-00-420-002

Refer to Figure 501.

A. Do the operational test of the No. 1 hydraulic system case drain filter manifold DPI and bypass valve as follows:

- (1) Get access to the No. 1 hydraulic system filter manifold (1) through the open left main landing gear door.
- (2) Do the procedure that follows to get the No. 1 hydraulic system to normal operating temperature:
  - (a) Pressurize the No. 1 hydraulic system (Refer to [TASK 29-00-00-863-801](#)).
  - (b) For the No. 1 hydraulic system, operate the rudder, elevators, flaps, and inboard spoilers until the system gets to full operating temperature.

**CAUTION:** DO NOT LET THE TEMPERATURE GET MORE THAN 225 °F (107 °C) AROUND THE DPI AND FILTER MANIFOLD. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE COMPONENTS.

- (c) If necessary, use a heat gun to heat the head (near the DPI) of the applicable case drain filter manifold to above 80 °F (26.66 °C).

**NOTE:** The DPI has a thermal lockout that will not let the red DPI indicator extend until it is heated above 80 °F (26.66 °C).

- (3) Remove hydraulic pressure from the No. 1 and No. 2 hydraulic system (Refer to [TASK 29-00-00-863-802](#)).
- (4) Put a hydraulic fluid container under the work area.



## AIRCRAFT MAINTENANCE MANUAL

**WARNING:** USE GOGGLES AND GLOVES WHEN YOU DO WORK WITH HYDRAULIC FLUID. IT CAN CAUSE IRRITATION TO YOUR EYES, SKIN, AND RESPIRATORY SYSTEM. IF IT GETS IN YOUR EYES, FLUSH YOUR EYES WITH WATER. IF IT TOUCHES YOUR SKIN, IMMEDIATELY CLEAN THE SKIN WITH SOAP AND WATER.

- (5) Remove the case drain filter bowl and filter element (Refer to [TASK 29-11-34-000-801](#)).
- (6) Disconnect the inlet line (2) from the case drain filter manifold (1).
- (7) Connect a hydraulic hand pump to the inlet fitting (3) on the case drain filter manifold (1).
- (8) Use the hydraulic hand pump to slowly increase the pressure to 15 psi (103 kPa) and stop. Keep the pressure on the case drain filter manifold for 1 minute. The case drain filter manifold DPI (4) should not extend.

**WARNING:** DO NOT TOUCH THE DPI OR THE AREA AROUND IT WITH YOUR BARE FINGER. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURY TO YOURSELF. MAKE SURE YOU USE HEAT PROTECTIVE GLOVES.

- (9) Slowly increase the pressure to 21.5 psi (148 kPa). The case drain filter manifold DPI (4) should extend. Do not push the DPI back into the case drain filter manifold housing.

**NOTE:** The red DPI indicator will extend approximately 0.188 in (0.48 cm).

**NOTE:** It can take approximately three seconds for the DPI to extend.

- (a) If the DPI (4) does not extend, do the tasks to replace the No. 1 hydraulic system filter manifold (Refer to [TASK 29-11-31-000-801](#) and [TASK 29-11-31-400-801](#)).
  - (b) If you replace the filter manifold, go to the Close Out section of this task.
- (10) Slowly increase the pressure to 85 psi (586 kPa) and hold it for 1 minute. The bypass valve (5) should not open.

- (a) If the pressure will not stay steady, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).

- (b) If you replace the bypass valve, go to the Close Out section of this task.

**NOTE:** If the pressure will not stay stable (or near stable) at 85 psi (586 kPa), then the bypass valve is in bypass mode at too low of a pressure and it is not serviceable.

- (11) Increase the pressure to 90 psi (621 kPa) and hold it for 1 minute. The bypass valve (5) should open. If pressure stays stable at 90 psi (621 kPa) for longer than one minute, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).

**NOTE:** You will know when the bypass valve opens, there will be a sudden pressure drop.

- (a) If the pressure will not stay steady or near steady, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).

- (b) If you replace the bypass valve, go to the Close Out section of this task.



## AIRCRAFT MAINTENANCE MANUAL

- (12) Release the pressure fully from the hydraulic hand pump.
- (13) Slowly increase the pressure to 85 psi (586 kPa) again and hold it for 1 minute.
  - (a) If the pressure will not stay steady, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).
  - (b) If the pressure will not stay steady or near steady or not build pressure, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).
  - (c) If you replace the bypass valve (5), go to the Close Out section of this task.
- (14) Release the pressure from the hydraulic hand pump.
- (15) Disconnect the hydraulic hand pump.
- (16) Install the No. 1 hydraulic system case drain manifold inlet line.
- (17) Torque the inlet line (2) (Refer to [TASK 20-20-11-400-801](#)).
- (18) Install the No. 1 hydraulic system case drain manifold filter element and filter bowl (Refer to [TASK 29-11-34-400-801](#)).
- (19) Do a quantity check of the No. 1 hydraulic system reservoir (Refer to [TASK 12-10-29-210-801](#)).
- (20) Service the No. 1 hydraulic system as necessary (Refer to [TASK 12-10-29-611-801](#)).
- (21) Bleed the No. 1 hydraulic system (Refer to [TASK 29-10-00-870-801](#)).

■ \*\*ON A/C 4001, 4003, 4006, 4008-4022, 4024-4029, 4033-4036, 4038-4045, 4052, 4056, 4058, 4063, 4065-4068, 4072-4080, 4082, 4084-4085, 4087-4114, 4116-4140, 4142, 4144, 4146, 4148, 4151-4152, 4155, 4157-4159, 4162, 4165, 4168, 4170-4172, 4174, 4176-4177, 4179-4199, 4201-4204, 4206-4209, 4212-4221, 4223-4226, 4228-4231, 4233-4234, 4237-4239, 4241-4242, 4245, 4247 Pre SB84-29-21

Subtask 29-10-00-420-003

[Refer to Figure 501.](#)

- B. Do the operational test of the No. 2 hydraulic system case drain filter manifold DPI and bypass valve as follows:
  - (1) Get access to the No. 2 hydraulic system filter manifold (1) through the open right main landing gear doors.
  - (2) Do the procedure that follows to get the No. 2 hydraulic system to normal operating temperature:
    - (a) Pressurize the No. 2 hydraulic system (Refer to [TASK 29-00-00-863-801](#)).
    - (b) For the No. 2 hydraulic system, operate the rudder, elevators, outboard spoilers, and emergency parking brake until the system gets to full operating temperature.

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-10-00



## AIRCRAFT MAINTENANCE MANUAL

**CAUTION:** DO NOT LET THE TEMPERATURE GET MORE THAN 225 °F (107 °C) AROUND THE DPI AND FILTER MANIFOLD. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE COMPONENTS.

- (c) If necessary, use a heat gun to heat the head (near the DPI) of the applicable case drain filter manifold to above 80 °F (26.66 °C).

**NOTE:** The DPI has a thermal lockout that will not let the red DPI indicator extend until it is heated above 80 °F (26.66 °C).

- (3) Remove hydraulic pressure from the No. 1 and No. 2 hydraulic system (Refer to [TASK 29-00-00-863-802](#)).
- (4) Put a hydraulic fluid container under the work area.

**WARNING:** USE GOGGLES AND GLOVES WHEN YOU DO WORK WITH HYDRAULIC FLUID. IT CAN CAUSE IRRITATION TO YOUR EYES, SKIN, AND RESPIRATORY SYSTEM. IF IT GETS IN YOUR EYES, FLUSH YOUR EYES WITH WATER. IF IT TOUCHES YOUR SKIN, IMMEDIATELY CLEAN THE SKIN WITH SOAP AND WATER.

- (5) Remove the case drain filter bowl and filter element (Refer to [TASK 29-11-34-000-801](#)).
- (6) Disconnect the inlet line (2) from the case drain filter manifold (1).
- (7) Connect a hydraulic hand pump to the inlet fitting (3) on the case drain filter manifold (1).
- (8) Use the hydraulic hand pump to slowly increase the pressure to 15 psi (103 kPa) and stop. Keep the pressure on the case drain filter manifold for 1 minute. The case drain filter manifold DPI (4) should not extend.

**WARNING:** DO NOT TOUCH THE DPI OR THE AREA AROUND IT WITH YOUR BARE FINGER. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURY TO YOURSELF. MAKE SURE YOU USE HEAT PROTECTIVE GLOVES.

- (9) Slowly increase the pressure to 21.5 psi (148 kPa). The case drain filter manifold DPI (4) should extend. Do not push the DPI back into the case drain filter manifold housing.

**NOTE:** The red DPI indicator will extend approximately 0.188 in (0.48 cm).

**NOTE:** It can take approximately three seconds for the DPI to extend.

- (a) If the DPI (4) does not extend, do the tasks to replace the No. 2 hydraulic system filter manifold (Refer to [TASK 29-11-31-000-801](#) and [TASK 29-11-31-400-801](#)).
- (b) If you replace the filter manifold, go to the Close Out section of this task.
- (10) Slowly increase the pressure to 85 psi (586 kPa) and hold it for 1 minute. The bypass valve (5) should not open.
- (a) If the pressure will not stay steady, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).



## AIRCRAFT MAINTENANCE MANUAL

- (b) If you replace the bypass valve, go to the Close Out section of this task.

**NOTE:** If the pressure will not stay stable (or near stable) at 85 psi (586 kPa), then the bypass valve is in bypass mode at too low of a pressure and it is not serviceable.

- (11) Increase the pressure to 90 psi (621 kPa) and hold it for 1 minute. The bypass valve (5) should open. If pressure stays stable at 90 psi (621 kPa) for longer than one minute, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).

**NOTE:** You will know when the bypass valve opens, there will be a sudden pressure drop.

- (a) If the pressure will not stay steady or near steady, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).

- (b) If you replace the bypass valve, go to the Close Out section of this task.

- (12) Release the pressure fully from the hydraulic hand pump.

- (13) Slowly increase the pressure to 85 psi (586 kPa) again and hold it for 1 minute.

- (a) If the pressure will not stay steady, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).

- (b) If the pressure will not stay steady or near steady or not build pressure, replace the bypass valve (Refer to [TASK 29-11-37-000-801](#) and [TASK 29-11-37-400-801](#)).

- (c) If you replace the bypass valve (5), go to the Close Out section of this task.

- (14) Release the pressure from the hydraulic hand pump.

- (15) Disconnect the hydraulic hand pump.

- (16) Install the No. 2 hydraulic system case drain manifold inlet line.

- (17) Torque the inlet line (2) (Refer to [TASK 20-20-11-400-801](#)).

- (18) Install the No. 2 hydraulic system case drain manifold filter element and filter bowl (Refer to [TASK 29-11-34-400-801](#)).

- (19) Do a quantity check of the No. 2 hydraulic system reservoir (Refer to [TASK 12-10-29-210-801](#)).

- (20) Service the No. 2 hydraulic system as necessary (Refer to [TASK 12-10-29-611-801](#)).

- (21) Bleed the No. 2 hydraulic system (Refer to [TASK 29-10-00-870-802](#)).

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## AIRCRAFT MAINTENANCE MANUAL

### 5. Close Out

Subtask 29-10-00-160-001

- A. Remove the tags and close the circuit breakers that follow:

| CB PANEL AND CB NO. | NAME           |
|---------------------|----------------|
| RIGHT DC, Q4        | STBY HYD PMP 1 |
| RIGHT DC, L8        | HYD PWR XFER   |

Subtask 29-10-00-941-057

- B. Clean all hydraulic fluid leakage from the area.
- C. Remove all tools, equipment, and unwanted materials from the work area.
- D. Dispose of all contaminated materials, wiper cloths and hydraulic fluid according to your Local Government Health and Safety Regulations.

Subtask 29-10-00-010-048

**WARNING:** BE CAREFUL WHEN YOU REMOVE THE DOOR LOCKPIN. THE LANDING GEAR DOORS WILL CLOSE IF YOU PRESSURIZE THE HYDRAULIC SYSTEM. IF YOU DO NOT DO THIS, IT CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

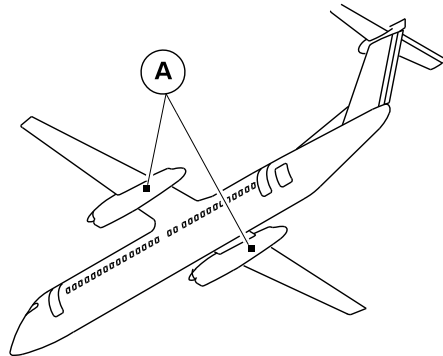
- E. Unlock and close the landing gear doors (Refer to [TASK 32-00-00-840-802](#)).





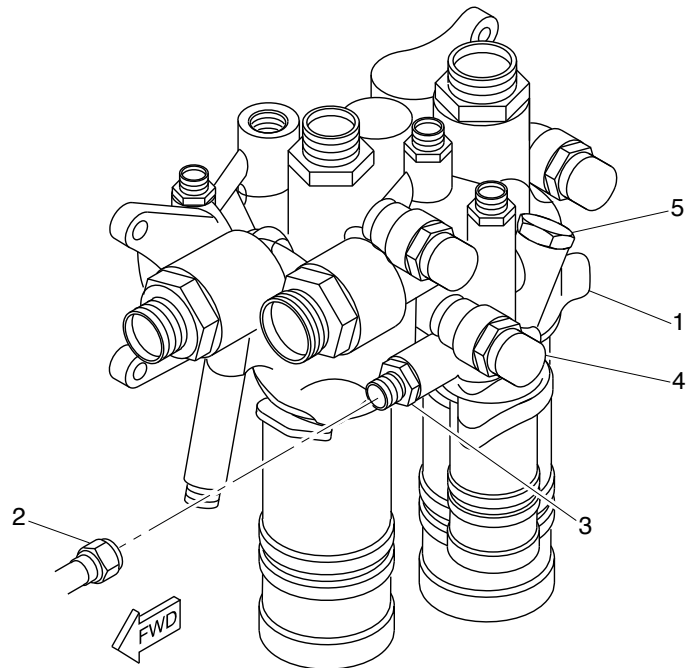
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## AIRCRAFT MAINTENANCE MANUAL



### LEGEND

1. Hydraulic system filter manifold.
2. Hydraulic line, case drain inlet.
3. Case drain inlet fitting.
4. Differential pressure indicator.
5. Bypass valve.



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Operational Test of the No. 1 and No. 2 Hydraulic System Filter Manifold Case Drain Bypass Valve  
Figure 501

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-10-00

29-10-00 Page 509  
Sep 05/2021



## AIRCRAFT MAINTENANCE MANUAL

### MAIN HYDRAULIC SYSTEMS 1 AND 2 – ADJUSTMENT/TEST

\*\*ON A/C ALL

#### **TASK 29-11-00-710-801**

#### **Operational Test of the No.1 and No. 2 Main Hydraulic Systems**

##### 1. General

- A. The maintenance procedure that follows is for the operational test of the No.1 and No. 2 Hydraulic Systems. No. 1 hydraulic system is installed in the left nacelle and center wing left side. No. 2 hydraulic system is installed in the right nacelle and center wing right side.

##### 2. Job Set-Up Information

Subtask 29-11-00-946-003

##### A. Reference Information

| REFERENCE                             | DESIGNATION                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------|
| <a href="#">TASK 29-00-00-910-801</a> | Hydraulic Safety Precautions                                                  |
| <a href="#">TASK 10-11-00-587-801</a> | Parking of the Aircraft.                                                      |
| <a href="#">TASK 12-10-29-210-801</a> | Fluid Quantity Check of the No. 1 and No. 2 Hydraulic Systems Reservoirs.     |
| <a href="#">TASK 12-10-29-611-801</a> | Servicing of the No. 1 and No. 2 Hydraulic Systems Reservoirs                 |
| <a href="#">TASK 24-00-00-861-801</a> | Energize the Electrical System.                                               |
| <a href="#">TASK 24-00-00-861-802</a> | De-energize the Electrical System.                                            |
| <a href="#">TASK 29-00-00-863-801</a> | Pressurization of No. 1 and No. 2 Hydraulic Systems.                          |
| <a href="#">TASK 29-00-00-863-802</a> | Removal of the Hydraulic Pressure from the No. 1 and No. 2 Hydraulic Systems. |
| <a href="#">TASK 32-31-00-710-801</a> | Operational test of the Landing Gear Primary Extension and Retraction System  |
| <a href="#">TASK 32-40-00-710-801</a> | Operational Check of the Emergency Park Brake Control Valve.                  |
| <a href="#">TASK 32-42-00-720-801</a> | Functional Test of the Main Landing Gear Brake System                         |
| <a href="#">TASK 32-44-00-720-801</a> | Functional Test of the Parking Brake System                                   |
| <a href="#">TASK 32-51-00-710-801</a> | Operational Test of the Nosewheel Steering System                             |
| <a href="#">TASK 27-30-00-710-801</a> | Operational Test of the Elevator Control System                               |
| <a href="#">TASK 27-50-00-710-801</a> | Operational Test of the Wing Flaps.                                           |
| <a href="#">TASK 27-61-00-710-801</a> | Operational Test of the Roll Spoilers System                                  |

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-11-00

**29-11-00** Page 501  
Sep 05/2021



**AIRCRAFT MAINTENANCE MANUAL**

**TASK 27-20-00-710-802**

Operational Test of the Rudder Control System

3. Job Set-Up:

Subtask 29-11-00-910-002

- A. Obey all hydraulic safety precautions. (Refer to [TASK 29-00-00-910-801](#)).

Subtask 29-11-00-780-001

- B. Pressurize the No. 1 and No. 2 Hydraulic Systems. (Refer to [TASK 29-00-00-863-801](#)).

4. Procedure

Subtask 29-11-00-400-001

- A. Do the Operational Test of the No. 1 or No. 2 Hydraulic System as follows

**WARNING:** OBEY ALL THE HYDRAULIC SAFETY PRECAUTIONS WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM OR A HYDRAULIC SYSTEM COMPONENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

**CAUTION:** BEFORE YOU SET THE BATTERY MASTER SWITCH TO ON POSITION, MAKE SURE THAT ALL THE MAINTENANCE WORK IS COMPLETE. IF YOU DO NOT DO THIS YOU MAY DAMAGE THE DCMP.

- (1) Make sure that the parking brake is set. (Refer to [TASK 10-11-00-587-801](#)).
- (2) Check the hydraulic fluid quantity in the No. 1 and No. 2 Hydraulic Systems Reservoirs. (Refer to [TASK 12-10-29-210-801](#)).
- (3) If necessary, service the No. 1 and No. 2 Hydraulic Systems Reservoirs with hydraulic fluid. (Refer to [TASK 12-10-29-611-801](#)).
- (4) Select to pressurize the No. 1 Hydraulic System by running No. 1 EDP to test No. 1 Hydraulic System. (Refer to [TASK 29-00-00-863-801](#)).
- (5) For No. 1 Hydraulic System, operate the flaps for 6 full cycles. (Refer to [TASK 27-50-00-710-801](#)).
- (6) For No. 1 Hydraulic System, operate the Spoilers for 6 full cycles. (Refer to [TASK 27-61-00-710-801](#)).
- (7) For No. 1 Hydraulic System, operate the Elevators for 6 full cycle. (Refer to [TASK 27-30-00-710-801](#)).
- (8) For No. 1 Hydraulic System, operate the Rudders for 6 full cycle. (Refer to [TASK 27-20-00-710-802](#)).

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-11-00

**29-11-00** Page 502  
Sep 05/2021



## AIRCRAFT MAINTENANCE MANUAL

- (9) For No. 1 Hydraulic System, operate the main wheel brakes for 6 full cycles. (Refer to [TASK 32-42-00-720-801](#)).
  - (10) Remove the pressure from the No. 1 Hydraulic System. (Refer to [TASK 29-00-00-863-802](#)).
  - (11) Select to pressurize the No. 2 Hydraulic System by running No. 2 EDP to test No. 2 Hydraulic System. (Refer to [TASK 29-00-00-863-801](#)).
  - (12) For No. 2 Hydraulic System, operate the Spoilers for 6 full cycles. (Refer to [TASK 27-61-00-710-801](#)).
  - (13) For No. 2 Hydraulic System, operate the Elevators for 6 full cycle. (Refer to [TASK 27-30-00-710-801](#)).
  - (14) For No. 2 Hydraulic System, operate the Rudders for 6 full cycle. (Refer to [TASK 27-20-00-710-802](#)).
  - (15) For No. 2 Hydraulic System, operate the Nose wheel steering for 6 full cycles. (Refer to [TASK 32-51-00-710-801](#)).
  - (16) For No. 2 Hydraulic System, operate the emergency parking brakes for 6 full cycle. (Refer to [TASK 32-40-00-710-801](#)).
  - (17) Remove the pressure from the No. 2 Hydraulic System. (Refer to [TASK 29-00-00-863-802](#)).
  - (18) Select to pressurize the No. 1 Hydraulic System by running No. 1 SPU to test No. 1 Hydraulic System. (Refer to [TASK 29-00-00-863-801](#)).
  - (19) Repeat steps number (5) to (9).
  - (20) Remove the pressure from the No. 1 Hydraulic System. (Refer to [TASK 29-00-00-863-802](#)).
  - (21) Select to pressurize the No. 2 Hydraulic System by running No. 1 SPU and PTU of the No. 2 Hydraulic System to test No. 2 Hydraulic System. (Refer to [TASK 29-00-00-863-801](#)).
  - (22) For No. 2 Hydraulic System, operate the landing gear extension and retraction system for 2 full cycles. (Refer to [TASK 32-31-00-710-801](#)).
- NOTE:** During Main Landing Gear extension the PTU pressure will fluctuate.
- (23) Repeat steps number (12) to (16).
  - (24) Remove the pressure from the No. 2 Hydraulic System. (Refer to [TASK 29-00-00-863-802](#)).
  - (25) De-energize the electrical system. (Refer to [TASK 24-00-00-861-802](#)).



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## AIRCRAFT MAINTENANCE MANUAL

### 5. Close Out

Subtask 29-11-00-941-004

- A. Remove all tools, equipment, and unwanted materials from the work area.



## AIRCRAFT MAINTENANCE MANUAL

### VALVE, HYDRAULIC — FIREWALL SHUT-OFF – ADJUSTMENT/TEST

\*\*ON A/C ALL

#### **TASK 29-11-06-710-801**

#### **Operational Test of the Firewall Shut-off Hydraulic Valve**

Refer to Figure 501.

##### 1. General

- A. The maintenance procedure that follows is for the operational test of the Firewall Shut-off Hydraulic Valve (written as the “shut-off valve” in this procedure). The No. 1 hydraulic system shut-off valve is installed in the left side nacelle, on the variable delivery hydraulic pump suction line. The No. 2 hydraulic system shut-off valve is installed in the right side nacelle, on the variable delivery hydraulic pump suction line. This procedure is applicable to the No. 1 and the No. 2 hydraulic systems. The differences are identified.

##### 2. Job Set-Up Information

Subtask 29-11-06-946-001

##### A. Reference Information

| REFERENCE                             | DESIGNATION                                                             |
|---------------------------------------|-------------------------------------------------------------------------|
| <a href="#">TASK 24-00-00-861-801</a> | Energize the Electrical System                                          |
| <a href="#">TASK 24-00-00-861-802</a> | De-energize the Electrical System                                       |
| <a href="#">TASK 24-00-00-910-801</a> | Electrical/Electronic Safety Precautions                                |
| <a href="#">TASK 12-10-29-210-801</a> | Fluid Quantity Check of the No. 1 and No. 2 Hydraulic System Reservoirs |

##### 3. Job Set-Up

**NOTE:** YOU CAN DO THE OPERATIONAL TEST OF THE FIREWALL SHUT-OFF HYDRAULIC VALVE BY USING EXTERNAL DC OR AC ELECTRICAL POWER TO ENERGIZE THE AIRCRAFT ELECTRICAL SYSTEMS.

Subtask 29-11-06-910-002

- A. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#) ).

Subtask 29-11-06-861-001

- B. Energize the electrical system (Refer to [TASK 24-00-00-861-801](#) ).



## AIRCRAFT MAINTENANCE MANUAL

Subtask 29-11-06-865-001

C. Open and tag the circuit breakers that follow:

| CB PANEL AND CB NO       | NAME                                     |
|--------------------------|------------------------------------------|
| RIGHT DC (ESSENTIAL), F2 | BATTERY PWR – FIRE EXT/CONT AMP 1& 2 ENG |
| RIGHT DC (ESSENTIAL), G2 | BATTERY PWR – FIRE EXT/CONT AMP CARGO    |

Subtask 29-11-06-865-005

D. Make sure that the circuit breakers that follow are closed:

| CB PANEL AND CB NO       | NAME                                                              |
|--------------------------|-------------------------------------------------------------------|
| RIGHT DC (ESSENTIAL), F1 | BATTERY PWR – HYD SOV – ENG 1 (For the left side – No. 1 system)  |
| RIGHT DC (ESSENTIAL), G1 | BATTERY PWR – HYD SOV – ENG 2 (For the right side – No. 2 system) |

Subtask 29-11-06-210-001

E. Do a quantity check of the No. 1 and No. 2 hydraulic system reservoirs (Refer to [TASK 12-10-29-210-801](#)).

**NOTE:** The hydraulic shut-off valves will close when the hydraulic reservoir fluid quantity is 0 to 0.075 US gal (0 to 0.28 L). This will cause the white HYD VALVES CLOSED indicator lights to come on.

#### 4. Procedure

Subtask 29-11-06-710-002

A. Do an operational test of the hydraulic shut-off valves as follows:

- (1) In the nacelles, visually check the shut-off valve position indicator. Make sure that the indicator arm shows the OPEN position.  
**NOTE:** Access the shut-off valves through the main landing gear doors.
- (2) On the flight compartment fire protection panel, pull the two PULL FUEL/HYD OFF fire handles.
- (3) In the nacelles, make sure that the shut-off valve indicator arm shows the CLOSED position.
- (4) In the flight compartment, make sure that the white HYD VALVES CLOSED indicator lights come on, and the green OPEN indicator lights go off.



## AIRCRAFT MAINTENANCE MANUAL

- (5) On the flight compartment fire protection panel, push the two PULL FUEL/HYD OFF fire handles in.
- (6) In the nacelles, make sure that the shut-off valve indicator arm shows the OPEN position.
- (7) In the flight compartment, make sure that the green HYD VALVES OPEN indicator lights come on, and the white CLOSED indicator lights go off.

### 5. Close Out

Subtask 29-11-06-861-002

- A. De-energize the electrical system (Refer to [TASK 24-00-00-861-802](#)).

Subtask 29-11-06-865-006

- B. Remove the tags and close the circuit breakers that follow:

| CB PANEL AND CB NO       | NAME                                     |
|--------------------------|------------------------------------------|
| RIGHT DC (ESSENTIAL), F2 | BATTERY PWR – FIRE EXT/CONT AMP 1& 2 ENG |
| RIGHT DC (ESSENTIAL), G2 | BATTERY PWR – FIRE EXT/CONT AMP CARGO    |

Subtask 29-11-06-941-002

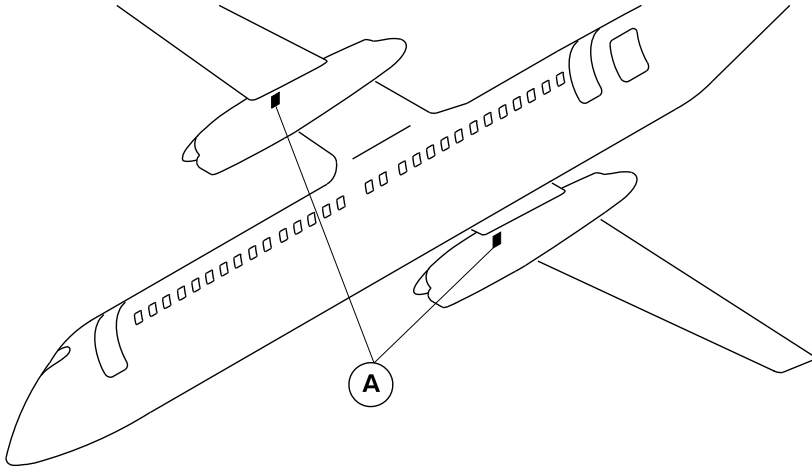
- C. Remove all tools, equipment, and unwanted materials from the work area.





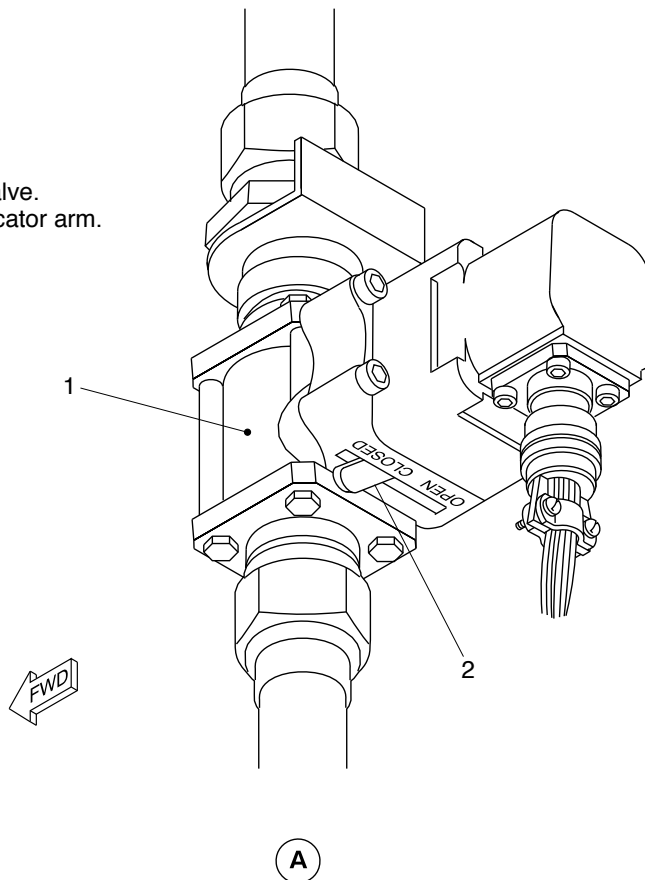
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## AIRCRAFT MAINTENANCE MANUAL



### LEGEND

1. Firewall shut-off valve.
2. FSOV position indicator arm.



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Operational Test of the Firewall Shut-off Hydraulic Valve  
Figure 501 (Sheet 1 of 2)

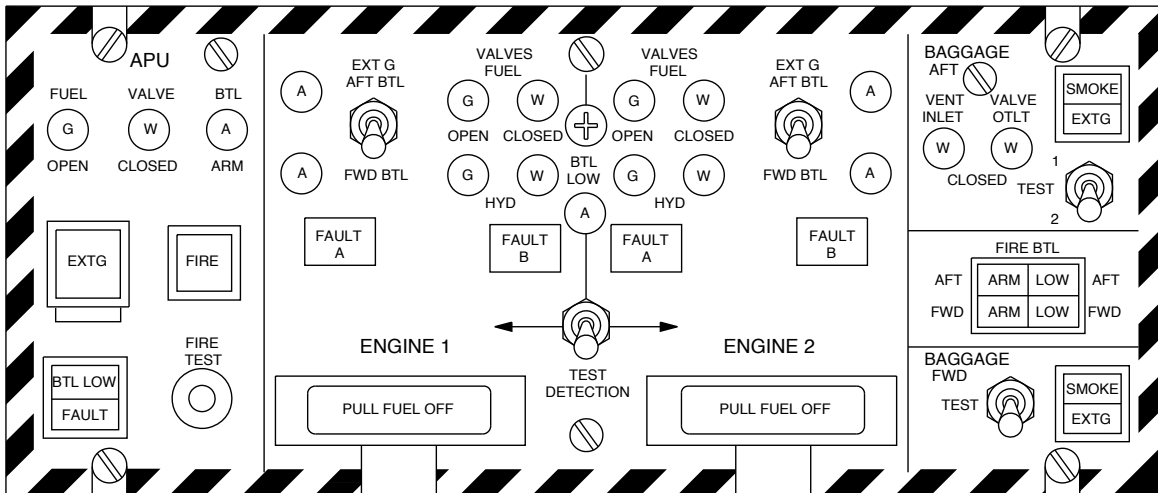
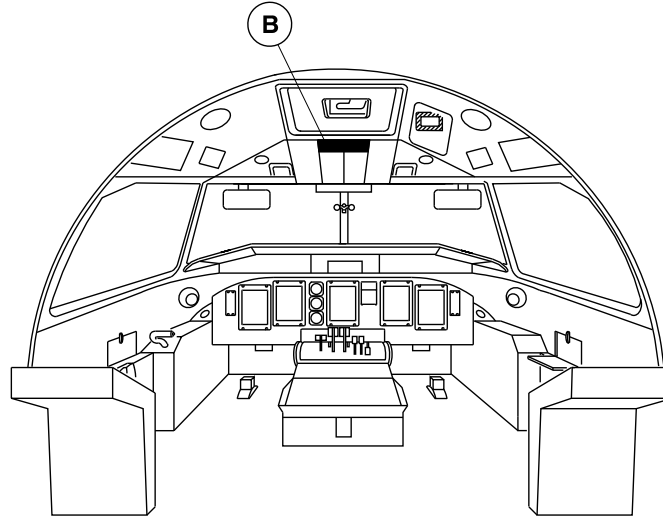
PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-11-06

29-11-06 Page 504  
Sep 05/2021



DE HAVILLAND AIRCRAFT  
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## AIRCRAFT MAINTENANCE MANUAL



### **B FIRE PROTECTION PANEL**

#### NOTE

The fire protection panel may vary for the aircraft with cargo combi configuration.

Operational Test of the Firewall Shut-off Hydraulic Valve  
Figure 501 (Sheet 2 of 2)

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-11-06

29-11-06 Page 505  
Sep 05/2021

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## AIRCRAFT MAINTENANCE MANUAL

### UNIT, STANDBY POWER — SYSTEM 1 – ADJUSTMENT/TEST

\*\*ON A/C ALL

#### **TASK 29-11-36-710-801**

#### **Operational Test of the Standby Power Unit**

##### 1. General

- A. The maintenance procedure that follows is for the operational test of the standby power unit (referred to as the "SPU" in this procedure). The SPU is installed in the left hand/No. 1 engine nacelle. This procedure is applicable to the No. 1 hydraulic system.

##### 2. Job Set-Up Information

Subtask 29-11-36-946-023

##### A. Reference Information

| REFERENCE                             | DESIGNATION                                                              |
|---------------------------------------|--------------------------------------------------------------------------|
| <a href="#">TASK 24-00-00-861-801</a> | Energize Electrical System                                               |
| <a href="#">TASK 24-00-00-861-802</a> | De-energize Electrical System                                            |
| <a href="#">TASK 24-00-00-910-801</a> | Electrical/Electronic Safety Precautions                                 |
| <a href="#">TASK 29-00-00-910-801</a> | Hydraulic Safety Precautions                                             |
| <a href="#">TASK 12-00-06-861-803</a> | Connection of the External AC Electrical Power                           |
| <a href="#">TASK 12-00-06-861-804</a> | Removal of the External AC Electrical Power                              |
| <a href="#">TASK 29-00-00-863-802</a> | Removal of the hydraulic pressure from No 1 and No. 2 hydraulic systems. |

##### 3. Job Set-Up

Subtask 29-11-36-910-023

- A. Obey all the hydraulic safety precautions (Refer to [TASK 29-00-00-910-801](#)).

Subtask 29-11-36-910-024

- B. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 29-11-36-865-021

- C. Make sure CB Q4(Right Main) is set to STBY HYD PUMP 1.



## AIRCRAFT MAINTENANCE MANUAL

Subtask 29-11-36-861-006

- D. Energize the electrical system (Refer to [TASK 24-00-00-861-801](#)).

### 4. Procedure

Subtask 29-11-36-710-022

**WARNING:** OBEY ALL THE HYDRAULIC SAFETY PRECAUTIONS WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM OR A HYDRAULIC SYSTEM COMPONENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

- A. Do the operational test of the SPU as follows:

- (1) Make sure that the parking brake is set.
- (2) Make sure that the flaps are set to 0 degrees.
- (3) In the flight compartment, monitor the STBY pressure on the multifunctional display. The STBY pressure should be 0 psi.
- (4) Release the parking brake.
- (5) Set the flaps to more than 0 degrees.
- (6) Make sure that the SPU operates as shown by a STBY pressure indication of 2750 to 2900 psi.
- (7) Set the flaps to 0 degrees.
- (8) Make sure that the SPU operation stops and the STBY pressure is 0 psi.

**NOTE:** Make sure after 8 seconds the Flap Power light on the caution warning light panel comes on.

- (9) Set the parking brake.
- (10) In the flight compartment, on the hydraulic control panel, turn the STBY HYD PRESS switch on.
- (11) Make sure that the SPU operates as shown by a STBY pressure indication of 2750 to 2900 psi.
- (12) Set the STBY HYD PRESS switch to normal, and make sure that the SPU operation stops.
- (13) Make sure that you removed the hydraulic pressure from No. 1 hydraulic system. (Refer to [TASK 29-00-00-863-802](#)).



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## AIRCRAFT MAINTENANCE MANUAL

### 5. Close Out

Subtask 29-11-36-861-005

- A. De-energize the electrical system (Refer to [TASK 24-00-00-861-802](#)).

Subtask 29-11-36-941-022

- B. Remove all tools, equipment, and unwanted materials from the work area.

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-11-36

**29-11-36** Page 503  
Sep 05/2021

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**AIRCRAFT MAINTENANCE MANUAL**

VALVE, BALANCED RELIEF — SYSTEM 1 — ADJUSTMENT/TEST

\*\*ON A/C ALL

**TASK 29-11-51-710-801**

**Operational Test of the No. 1 Hydraulic System Balanced Relief Valve**

1. General

- A. The maintenance procedure that follows is for the operational test of the No. 1 hydraulic system balanced relief valve (referred to as the “priority valve” in this procedure). The priority valve is installed in the left side of the fuselage-to-wing fairing. This procedure is applicable to the No. 1 hydraulic system.

2. Job Set-Up Information

Subtask 29-11-51-943-001

A. Tools & Equipment

- |     |            |                                    |
|-----|------------|------------------------------------|
| (1) | GSB29000XX | Ground Hydraulic Power Unit (GHPU) |
| (2) | GSB2900002 | Hydraulic Coupling Kit             |

Subtask 29-11-51-946-001

B. Reference Information

| REFERENCE                             | DESIGNATION                                                      |
|---------------------------------------|------------------------------------------------------------------|
| <a href="#">TASK 24-00-00-861-801</a> | Energize the Electrical System                                   |
| <a href="#">TASK 24-00-00-861-802</a> | De-energize the Electrical System                                |
| <a href="#">TASK 12-00-16-490-801</a> | Connection of the Ground Hydraulic Power                         |
| <a href="#">TASK 12-00-16-090-801</a> | Removal of the Ground Hydraulic Power                            |
| <a href="#">TASK 12-10-29-611-801</a> | Servicing of the No. 1 and No. 2 Hydraulic System Reservoirs     |
| <a href="#">TASK 24-00-00-910-801</a> | Electrical/Electronic Safety Precautions                         |
| <a href="#">TASK 29-00-00-910-801</a> | Hydraulic Safety Precautions                                     |
| <a href="#">TASK 29-11-51-000-801</a> | Removal of the No. 1 Hydraulic System Balanced Relief Valve      |
| <a href="#">TASK 29-11-51-400-801</a> | Installation of the No. 1 Hydraulic System Balanced Relief Valve |





## AIRCRAFT MAINTENANCE MANUAL

### 3. Job Set-Up

Subtask 29-11-51-910-001

- A. Obey all the hydraulic safety precautions (Refer to [TASK 29-00-00-910-801](#)).

Subtask 29-11-51-910-002

- B. Obey all the electrical/electronic safety precautions (Refer to [TASK 24-00-00-910-801](#)).

Subtask 29-11-51-861-001

- C. Connect and energize the external DC power (Refer to [TASK 24-00-00-861-801](#)).

Subtask 29-11-51-865-001

- D. Open and tag the circuit breaker that follows:

| CB PANEL AND CB NO | NAME           |
|--------------------|----------------|
| RIGHT DC, Q4       | STBY HYD PMP 1 |

Subtask 29-11-51-010-001

- E. Open the access door that follows:

| REFERENCE | DESIGNATION                        |
|-----------|------------------------------------|
| 413CR     | Door, Hydraulics, Ground Servicing |



## AIRCRAFT MAINTENANCE MANUAL

### 4. Procedure

Subtask 29-11-51-710-001

**WARNING:** INSTALL THE SAFETY DEVICES, THE WARNING SIGNS, AND THE WARNING PLACARDS BEFORE YOU START A PROCEDURE ON OR AROUND:

- FLIGHT CONTROLS
- FLIGHT CONTROL SURFACES
- COMPONENTS THAT MOVE

MAKE SURE THAT THERE ARE NO PERSONS OR EQUIPMENT ON OR NEAR THE SURFACES OF THE FLIGHT CONTROLS. ACCIDENTAL MOVEMENT OF THE FLIGHT CONTROLS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

A. Do an operational test of the priority valve as follows:

- (1) Connect a hydraulic power unit to the No. 1 hydraulic system quick disconnects (Refer to [TASK 12-00-16-490-801](#)).
- (2) Start the hydraulic power unit. Set the unit to the "OPEN LOOP" mode.
- (3) Set the ground hydraulic power unit pressure to approximately 3000 psig (20684 kPa).
- (4) Continuously move the elevator and operate the flaps while an assistant monitors the control surface movements.
- (5) Slowly decrease the pressure of the hydraulic power unit.
- (6) When you get to 1750 psig (12066 kPa), the flap movement will stop. The elevators will continue to move.

**NOTE:** Flap movement can get slow at approximately 2400 psig (16547 kPa) before it stops.

**NOTE:** The FLAP POWER caution light will illuminate and the FLAP DRIVE caution light could possibly illuminate also.

- (7) Make sure the No. 1 HYD ISO VLV caution light does not come on.
- (8) Replace the priority valve if it does not operate correctly (Refer to [TASK 29-11-51-000-801](#) and [TASK 29-11-51-400-801](#)).
- (9) Increase the pressure of the hydraulic power unit to 3000 psig (20684 kPa).
- (10) To extinguish the FLAP POWER and FLAP DRIVE caution lights, do the steps that follow:
  - (a) Open the following circuit breaker:

| CB PANEL AND CB NO | NAME       |
|--------------------|------------|
| Left DC, L7        | FLAPS CONT |

- (b) Wait approximately 10 seconds.

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-11-51



## AIRCRAFT MAINTENANCE MANUAL

(c) Close the following circuit breaker:

| CB PANEL AND CB NO | FLAPS CONT |
|--------------------|------------|
| Left DC, L7        | FLAPS CONT |

**NOTE:** It may take up to 30 seconds after you close the FLAPS CONT circuit breaker for the FLAP POWER and FLAP DRIVE caution lights to extinguish.

- (11) Make sure that the elevators move and the flaps fully operate.
- (12) Stop the movement of the flight controls. Turn off the hydraulic power unit.
- (13) Disconnect the hydraulic power unit from the hydraulic system quick disconnects (Refer to [TASK 12-00-16-090-801](#)).
- (14) Do the servicing of the No. 1 hydraulic system reservoir (Refer to [TASK 12-10-29-611-801](#)).

### 5. Close Out

Subtask 29-11-51-861-002

- A. Remove the external DC power (Refer to [TASK 24-00-00-861-802](#)).

Subtask 29-11-51-865-002

- B. Remove the tag and close the circuit breaker that follows:

| CB PANEL AND CB NO | NAME           |
|--------------------|----------------|
| RIGHT DC, Q4       | STBY HYD PMP 1 |

Subtask 29-11-51-410-001

- C. Close the access door that follows:

| REFERENCE | DESIGNATION                        |
|-----------|------------------------------------|
| 413CR     | Door, Hydraulics, Ground Servicing |

Subtask 29-11-51-941-004

- D. Remove all tools, equipment, and unwanted materials from the work area.



## AIRCRAFT MAINTENANCE MANUAL

### UNIT, POWER TRANSFER — HYDRAULIC — ADJUSTMENT/TEST

\*\*ON A/C ALL

#### **TASK 29-11-61-710-801**

#### **Operational Test of the Power Transfer Unit**

##### 1. General

- A. The maintenance procedure that follows is for the operational test of the power transfer unit. This procedure is applicable only to the No. 1 and No. 2 hydraulic systems.

##### 2. Job Set-Up Information

Subtask 29-11-61-943-002

##### A. Tools & Equipment

- |     |            |                        |
|-----|------------|------------------------|
| (1) | GSB2900001 | Hydraulic Power Unit   |
| (2) | GSB2900002 | Hydraulic Coupling Kit |

Subtask 29-11-61-946-003

##### B. Reference Information

| REFERENCE                             | DESIGNATION                                                             |
|---------------------------------------|-------------------------------------------------------------------------|
| <a href="#">TASK 12-00-06-861-801</a> | Connection of the External DC Electrical Power                          |
| <a href="#">TASK 12-00-06-861-802</a> | Removal of the External DC Electrical Power                             |
| <a href="#">TASK 12-00-06-861-803</a> | Connection of the External AC Electrical Power                          |
| <a href="#">TASK 12-00-06-861-804</a> | Removal of the External AC Electrical Power                             |
| <a href="#">TASK 12-00-06-861-805</a> | Connection of the External AC Electrical Power (Forward Fuselage)       |
| <a href="#">TASK 12-00-06-861-806</a> | Removal of the External AC Electrical Power (Forward Fuselage)          |
| <a href="#">TASK 12-00-11-861-801</a> | Energize the Electrical System with Battery Power.                      |
| <a href="#">TASK 12-00-11-861-802</a> | De-energize the Electrical System with Battery Power.                   |
| <a href="#">TASK 12-10-29-210-801</a> | Fluid Quantity Check of the No. 1 and No. 2 Hydraulic System Reservoirs |
| <a href="#">TASK 12-10-29-611-801</a> | Servicing of the No. 1 and No. 2 Hydraulic Systems Reservoirs           |
| <a href="#">TASK 12-10-29-210-802</a> | Fluid Quantity Check of the No. 3 Hydraulic System Reservoir            |
| <a href="#">TASK 12-10-29-611-802</a> | Servicing of the No. 3 Hydraulic System Reservoir                       |

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-11-61

**29-11-61** Page 501  
Sep 05/2021



**AIRCRAFT MAINTENANCE MANUAL**

| REFERENCE                             | DESIGNATION                                                         |
|---------------------------------------|---------------------------------------------------------------------|
| <a href="#">TASK 29-00-00-863-801</a> | Pressurization of the No. 1 and No. 2 Hydraulic Systems.            |
| <a href="#">TASK 29-00-00-863-802</a> | Removal of the pressure from the No. 1 and No. 2 Hydraulic Systems. |
| <a href="#">TASK 29-00-00-863-803</a> | Pressurization of the No. 3 Hydraulic System.                       |
| <a href="#">TASK 29-00-00-863-804</a> | Removal of the pressure from the No. 3 Hydraulic System.            |

3. Job Set-Up

Subtask 29-11-61-860-004

- A. Do a fluid quantity check of the No. 1 and No. 2 Hydraulic System Reservoirs (Refer to [TASK 12-10-29-210-801](#)).

Subtask 29-11-61-944-002

- B. If necessary, service the No. 1 and No. 2 Hydraulic Systems Reservoirs. Ref [TASK 12-10-29-611-801](#)

Subtask 29-11-61-861-001

- C. Connect the External DC Electrical Power (Refer to [TASK 12-00-06-861-801](#) ).

Subtask 29-11-61-861-002

- D. Connect the External AC Power (Refer to [TASK 12-00-06-861-803](#).)

Subtask 29-11-61-861-005

- E. Energize the Electrical System with Battery Power (Refer to [TASK 12-00-11-861-801](#) ).



## AIRCRAFT MAINTENANCE MANUAL

### 4. Procedure

**\*\*ON A/C 4001, 4003, 4006, 4008–4021, 4024–4029, 4033, 4035–4036, 4038–4045, 4052, 4054, 4056, 4058, 4064–4068, 4072–4080, 4082, 4084–4085, 4087–4140, 4142, 4144, 4146, 4148, 4151–4152, 4155, 4157–4159, 4162, 4165, 4168, 4170–4172, 4174, 4176–4177, 4179–4184 Pre SB84–29–22**

Subtask 29–11–61–710–001

#### A. Do the operational test of the PTU as follows:

- (1) Make sure the parking brake is set.
- (2) Make sure the flaps are set to 0 degrees.
- (3) Pressurize the No. 1 hydraulic system using the hydraulic power unit set to 3000 psi with a flow rate of 5 gpm (Refer to [TASK 29–00–00–863–801](#)).

**NOTE:** If a hydraulic power unit is not available, the standby power unit can be used to pressurize the No. 1 system.

- (4) In the flight compartment, set the PTU CNTRL switch to ON.
- (5) In the flight compartment, set the PTU CNTRL switch to NORMAL.
- (6) Remove hydraulic pressure from the No. 1 hydraulic system (Refer to [TASK 29–00–00–863–802](#)).
- (7) If the SPU was used to pressurize the No. 1 system, set the STBY HYD PRESS switch to NORMAL.

**\*\*ON A/C 4185–4199, 4201–4999 and ON A/C 4001, 4003–4004, 4006, 4008–4140, 4142–4184 Post SB84–29–22**

Subtask 29–11–61–710–002

#### B. Do the operational test of the PTU as follows:

- (1) Make sure the parking brake is set.
- (2) Make sure the flaps are set to 0 degrees.
- (3) Pressurize the No. 1 hydraulic system using the hydraulic power unit set to 3000 psi with a flow rate of 5 gpm (Refer to [TASK 29–00–00–863–801](#)).

**NOTE:** If a hydraulic power unit is not available, the standby power unit can be used to pressurize the No. 1 system.

- (4) In the flight compartment, set the PTU CNTRL switch to ON.
- (5) On the multi-function display, make sure that the No. 2 hydraulic system pressure is approximately 2700 psi. Make sure the “//” symbol illuminates on the PTU CNTRL switch.
- (6) In the flight compartment, set the PTU CNTRL switch to NORMAL.
- (7) Remove hydraulic pressure from the No. 1 hydraulic system (Refer to [TASK 29–00–00–863–802](#)).

PSM 1–84–2  
EFFECTIVITY:  
See first effectivity on page 501 of 29–11–61



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## AIRCRAFT MAINTENANCE MANUAL

- (8) If the SPU was used to pressurize the No. 1 system, set the STBY HYD PRESS switch to NORMAL.

\*\*ON A/C ALL

### 5. Close Out

Subtask 29-11-61-861-003

- A. Remove the external AC power (Refer to [TASK 12-00-06-861-804](#)).

Subtask 29-11-61-861-004

- B. Remove the external DC power (Refer to [TASK 12-00-06-861-802](#)).

Subtask 29-11-61-861-007

- C. De-energize the Electrical System with Battery Power (Refer to [TASK 12-00-11-861-802](#) ).

Subtask 29-11-61-941-002

- D. Remove all tools, equipment, and unwanted materials from the work area.



**AIRCRAFT MAINTENANCE MANUAL**

**PUMP, DC MOTOR DRIVEN — HYDRAULIC – ADJUSTMENT/TEST**

**\*\*ON A/C ALL**

**TASK 29-12-01-710-801**

**Operational Test of the No. 3 Hydraulic System after replacement of the DC Motor Driven Pump (DCMP).**

1. General

- A. The maintenance procedure that follows is for the Operational Test of the No. 3 Hydraulic System after replacement of the DC Motor Driven Pump (DCMP). The DCMP is installed in the aft fuselage. This procedure is applicable to the No. 3 hydraulic system.

This procedure is applicable to prove the serviceability of the systems that follow:

- (1) No. 3 hydraulic system.
- (2) No. 3 hydraulic system DCMP Pump.
- (3) No. 3 hydraulic system Accumulator Isolation Valve.
- (4) No. 3 hydraulic system Filter Manifold.
- (5) No. 3 hydraulic system 50 cu inch (819 cu cm) Accumulator.
- (6) No. 3 hydraulic system 25 cu inch (410 cu cm) Bootstrap Accumulator
- (7) No. 3 hydraulic system Reservoir Assembly.

2. Job Set-Up Information

Subtask 29-12-01-946-007

A. Reference Information

| REFERENCE                             | DESIGNATION                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------|
| <a href="#">TASK 29-00-00-910-801</a> | Hydraulic Safety Precautions                                                |
| <a href="#">TASK 10-11-00-587-801</a> | Parking of the Aircraft                                                     |
| <a href="#">TASK 24-00-00-861-801</a> | Energize the Electrical System                                              |
| <a href="#">TASK 24-00-00-861-802</a> | De-energize the Electrical System                                           |
| <a href="#">TASK 12-10-29-210-802</a> | Fluid Quantity Check of the No. 3 Hydraulic System Reservoir                |
| <a href="#">TASK 12-10-29-611-802</a> | Servicing of the No. 3 Hydraulic System Reservoir                           |
| <a href="#">TASK 12-10-29-614-801</a> | Servicing of the No. 3 Hydraulic System Accumulator with Nitrogen           |
| <a href="#">TASK 12-10-29-614-802</a> | Servicing of the No. 3 Hydraulic System Bootstrap Accumulator with Nitrogen |
| <a href="#">TASK 29-10-00-870-803</a> | Bleeding of the No. 3 Hydraulic System                                      |
| <a href="#">TASK 27-30-00-710-801</a> | Operational Test of the Elevator Control System                             |

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-12-01

**29-12-01** Page 501  
Sep 05/2021





**AIRCRAFT MAINTENANCE MANUAL**

| REFERENCE                             | DESIGNATION                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------|
| <a href="#">TASK 29-00-00-863-804</a> | Removal of the Hydraulic Pressure from the No. 3 Hydraulic System                |
| <a href="#">TASK 29-10-00-870-804</a> | Priming of the No. 3 System DCMP and Extended Bleeding of No. 3 Hydraulic System |

3. Job Set-Up

Subtask 29-12-01-865-012

- A. Make sure to close the circuit breakers that follow:

| CB PANEL AND CB NO       | NAME           |
|--------------------------|----------------|
| RIGHT DC (ESSENTIAL), M7 | HYD SYS 3 CONT |
| LEFT DC (ESSENTIAL), M9  | STBY BATT      |

Subtask 29-12-01-010-008

- B. Open the access door that follows:

| REFERENCE | DESIGNATION              |
|-----------|--------------------------|
| 311AB     | Aft Fuselage Access Door |

4. Procedure

\*\*ON A/C 4001, 4003-4004, 4006, 4008-4140, 4142-4199, 4201-4510, 4545

Subtask 29-12-01-710-002

- A. Do the Operational Test of the No. 3 Hydraulic System after replacement of the DC Motor Driven Pump (DCMP) as follows:



## AIRCRAFT MAINTENANCE MANUAL

**WARNING:** OBEY ALL THE HYDRAULIC SAFETY PRECAUTIONS WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM OR A HYDRAULIC SYSTEM COMPONENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

**CAUTION:** DO NOT OPERATE THE NUMBER 3 HYDRAULIC SYSTEM IF THE HYDRAULIC RESERVOIR HAS LESS THAN 45% OF THE FLUID QUANTITY WITH NO SYSTEM PRESSURE. IF YOU DO THIS YOU CAN CAUSE DAMAGE TO THE HYDRAULIC SYSTEM.

**CAUTION:** BEFORE YOU SET THE BATTERY MASTER SWITCH TO ON POSITION, MAKE SURE THAT ALL THE MAINTENANCE WORK IS COMPLETE. IF YOU DO NOT DO THIS YOU MAY DAMAGE THE DCMP.

- (1) Obey all hydraulic safety precautions (Refer to [TASK 29-00-00-910-801](#)).
- (2) Make sure that the parking brake is set (Refer to [TASK 10-11-00-587-801](#)).
- (3) Check the hydraulic fluid quantity in the No. 3 Hydraulic System Reservoir (Refer to [TASK 12-10-29-210-802](#)).
- (4) If necessary, service the No. 3 Hydraulic System Reservoir with hydraulic fluid (Refer to [TASK 12-10-29-611-802](#)).
- (5) If necessary, service the No. 3 Hydraulic System Accumulator with Nitrogen (Refer to [TASK 12-10-29-614-801](#)).
- (6) If necessary, service the No. 3 Hydraulic System Bootstrap Accumulator with Nitrogen (Refer to [TASK 12-10-29-614-802](#)).

**WARNING:** MAKE SURE YOU TELL ALL PERSONS IN THE AREA OF THE AIRCRAFT THAT YOU WILL ENERGIZE THE AIRCRAFT ELECTRICAL SYSTEM, BEFORE YOU TURN ON THE ELECTRICAL SYSTEM. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

- (7) Energize the electrical system to run the DCMP. (Refer to [TASK 24-00-00-861-801](#)).
- (8) Set the No. 3 Hydraulic System Isolation Valve "HYD # 3 ISOL VLV" in the cockpit hydraulic panel to "OPEN" position.
  - (a) DCMP must start.
  - (b) If the DCMP does not make pressure when it starts, stop the procedure and do the priming the No. 3 System DCMP. (Refer to [TASK 29-10-00-870-804](#)).
- (9) Operate the Elevators at a slow rate, until the No. 3 Hydraulic System pressure decreases to minimum of 2600 psi (17926 kPa) indicated on the hydraulic pressure display in the cockpit (Refer to [TASK 27-30-00-710-801](#)).
  - (a) DCMP must pressurize the No.3 Hydraulic System to 3000 psi (20684 kPa).
  - (b) If DCMP does not produce pressure of 3000 psi (20684 kPa) within 3 minutes, stop the procedure and bleed the No. 3 Hydraulic System. (Refer to [TASK 29-10-00-870-803](#) and [TASK 29-10-00-870-804](#)).



## AIRCRAFT MAINTENANCE MANUAL

- (c) The No. 3 Hydraulic System must operate smoothly and with normal DCMP pump sound free of chattering or banging.
  - (d) On the ESID Control Panel (ESCP) set the MFD1 to "NAV" mode and MFD2 to "SYS" mode.
  - (e) On the ESID Control Panel (ESCP) set the MFD1 to "PFD" mode and MFD2 to "SYS" mode.
- (10) Make sure that the DCMP recharges the accumulator of the No. 3 hydraulic system to 3000 psi (20684 kPa).
- (a) The DCMP pump starts at minimum of 2500 psi (17237 kPa) and stops at maximum of 3350 psi (23097 kPa).
- (11) Repeat the test by moving the elevators till the DCMP pump starts and stops two times.
- (12) Set the No. 3 Hydraulic System Isolation Valve "HYD # 3 ISOL VLV" in the cockpit hydraulic panel to "NORMAL" position to stop the DCMP pump.
- (a) DCMP must stop.
  - (b) Cycle the elevator and note that the No. 3 Hydraulic System pressure indication in the cockpit decreases.
- (13) Make sure that the No. 3 Hydraulic System pressure decreases to less than 1000 psi (6895 kPa) on the pressure indication in the cockpit.
- (a) On the ESID Control Panel (ESCP) set the MFD1 to "NAV" mode and MFD2 to "SYS" mode.
  - (b) On the ESID Control Panel (ESCP) set the MFD1 to "PFD" mode and MFD2 to "SYS" mode.
- (14) Check the hydraulic fluid quantity in the No. 3 Hydraulic System Reservoir (Refer to [TASK 12-10-29-210-802](#)).
- (15) If necessary, service the No. 3 Hydraulic System Reservoir with hydraulic fluid (Refer to [TASK 12-10-29-611-802](#)).
- (16) Remove the pressure from the No. 3 hydraulic system (Refer to [TASK 29-00-00-863-804](#)).
- (17) De-energize the electrical system (Refer to [TASK 24-00-00-861-802](#)).

\*\*ON A/C 4511-4544, 4546-4999

Subtask 29-12-01-710-003

- B. Do the Operational Test of the No. 3 Hydraulic System after replacement of the DC Motor Driven Pump (DCMP) as follows:

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-12-01

29-12-01 Page 504  
Sep 05/2021



## AIRCRAFT MAINTENANCE MANUAL

**WARNING:** OBEY ALL THE HYDRAULIC SAFETY PRECAUTIONS WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM OR A HYDRAULIC SYSTEM COMPONENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

**CAUTION:** DO NOT OPERATE THE NUMBER 3 HYDRAULIC SYSTEM IF THE HYDRAULIC RESERVOIR HAS LESS THAN 45% OF THE FLUID QUANTITY WITH NO SYSTEM PRESSURE. IF YOU DO THIS YOU CAN CAUSE DAMAGE TO THE HYDRAULIC SYSTEM.

**CAUTION:** BEFORE YOU SET THE BATTERY MASTER SWITCH TO ON POSITION, MAKE SURE THAT ALL THE MAINTENANCE WORK IS COMPLETE. IF YOU DO NOT DO THIS YOU MAY DAMAGE THE DCMP.

- (1) Obey all hydraulic safety precautions (Refer to [TASK 29-00-00-910-801](#)).
- (2) Make sure that the parking brake is set (Refer to [TASK 10-11-00-587-801](#)).
- (3) Check the hydraulic fluid quantity in the No. 3 Hydraulic System Reservoir (Refer to [TASK 12-10-29-210-802](#)).
- (4) If necessary, service the No. 3 Hydraulic System Reservoir with hydraulic fluid (Refer to [TASK 12-10-29-611-802](#)).
- (5) If necessary, service the No. 3 Hydraulic System Accumulator with Nitrogen (Refer to [TASK 12-10-29-614-801](#)).

**WARNING:** MAKE SURE YOU TELL ALL PERSONS IN THE AREA OF THE AIRCRAFT THAT YOU WILL ENERGIZE THE AIRCRAFT ELECTRICAL SYSTEM, BEFORE YOU TURN ON THE ELECTRICAL SYSTEM. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

- (6) Energize the electrical system to run the DCMP (Refer to [TASK 24-00-00-861-801](#)).
- (7) Set the No.3 Hydraulic System Isolation Valve "HYD # 3 ISOL VLV" in the cockpit hydraulic panel to "OPEN" position.
  - (a) DCMP must start.
- (8) Operate the Elevators at a slow rate, until the No.3 Hydraulic System pressure decreases to minimum of 2600 psi (17926 kPa) indicated on the hydraulic pressure display in the cockpit (Refer to [TASK 27-30-00-710-801](#)).
  - (a) DCMP must pressurize the No.3 Hydraulic System to 3000 psi (20684 kPa).
  - (b) If DCMP does not produce pressure of 3000 psi (20684 kPa) within 3 minutes, stop the procedure and bleed the No. 3 Hydraulic System (Refer to [TASK 29-10-00-870-803](#) and [TASK 29-10-00-870-804](#)).
  - (c) The No. 3 Hydraulic System must operate smoothly and with normal DCMP pump sound free of chattering or banging.
  - (d) On the ESID Control Panel (ESCP) set the MFD1 to "NAV" mode and MFD2 to "SYS" mode.



## AIRCRAFT MAINTENANCE MANUAL

- (e) On the ESID Control Panel (ESCP) set the MFD1 to "PFD" mode and MFD2 to "SYS" mode.
- (9) Make sure that the DCMP recharges the accumulator of the No. 3 hydraulic system to 3000 psi (20684 kPa).
  - (a) The DCMP pump starts at minimum of 2500 psi (17237 kPa) and stops at maximum of 3350 psi (23097 kPa).
- (10) Repeat the test by moving the elevators till the DCMP pump starts and stops two times.
- (11) Set the No.3 Hydraulic System Isolation Valve "HYD # 3 ISOL VLV" in the cockpit hydraulic panel to "NORMAL" position to stop the DCMP pump.
  - (a) DCMP must stop.
  - (b) Cycle the elevator and note that the No.3 Hydraulic System pressure indication in the cockpit decreases.
- (12) Make sure that the No.3 Hydraulic System pressure decreases to less than 1000 psi (6895 kPa) on the pressure indication in the cockpit.
  - (a) On the ESID Control Panel (ESCP) set the MFD1 to "NAV" mode and MFD2 to "SYS" mode.
  - (b) On the ESID Control Panel (ESCP) set the MFD1 to "PFD" mode and MFD2 to "SYS" mode.
- (13) Check the hydraulic fluid quantity in the No. 3 Hydraulic System Reservoir (Refer to [TASK 12-10-29-210-802](#)).
- (14) If necessary, service the No. 3 Hydraulic System Reservoir with hydraulic fluid (Refer to [TASK 12-10-29-611-802](#)).
- (15) Remove the pressure from the No. 3 hydraulic system (Refer to [TASK 29-00-00-863-804](#)).
- (16) De-energize the electrical system (Refer to [TASK 24-00-00-861-802](#)).

\*\*ON A/C ALL

### 5. Close Out

Subtask 29-12-01-410-007

A. Close the access door that follows:

| REFERENCE | DESIGNATION              |
|-----------|--------------------------|
| 311AB     | Aft Fuselage Access Door |



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## AIRCRAFT MAINTENANCE MANUAL

Subtask 29-12-01-941-007

- B. Remove all tools, equipment, and unwanted materials from the work area.

PSM 1-84-2  
EFFECTIVITY:  
See first effectivity on page 501 of 29-12-01

**29-12-01** Page 507  
Sep 05/2021

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